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Project 1

ECEN 3400

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Van de Graaff Generator – Project 1

Background

The Van de Graaff is an electrostatic generator that can produce extremely high voltages at the cost of an exceptionally low current. It was invented in 1929 by Robert J. Van de Graaff initially for his nuclear physics research. It functions by using a moving belt that transfers electric charge. It is constructed using a metal hollow sphere, usually of copper, aluminum, or steel, an insulated tube, and metal brushes near the moving belt. The charge comes from the moving belt system (usually a motor and a large band on tabletop setups). The triboelectric effect is utilized on the lower comb to transfer charge to the belt. The charge gets transported up to the metal brush inside the sphere where it gets deposited on the surface. Since the sphere is a conductor, the charge gets constantly distributed slowly increasing its potential.

Even at smaller scales, Van de Graaff generators can create a massive amount of voltage. The largest generator, created by Van de Graaff himself, can produce up to five million volts. It has evolved since its invention. The transportation charge has gotten more efficient and preventing the loss of charge has improved. However, in the modern era, especially for experiments needing high energy levels and power, particle accelerators have largely replaced these Van de Graaff generators. The general structure looks like the following:

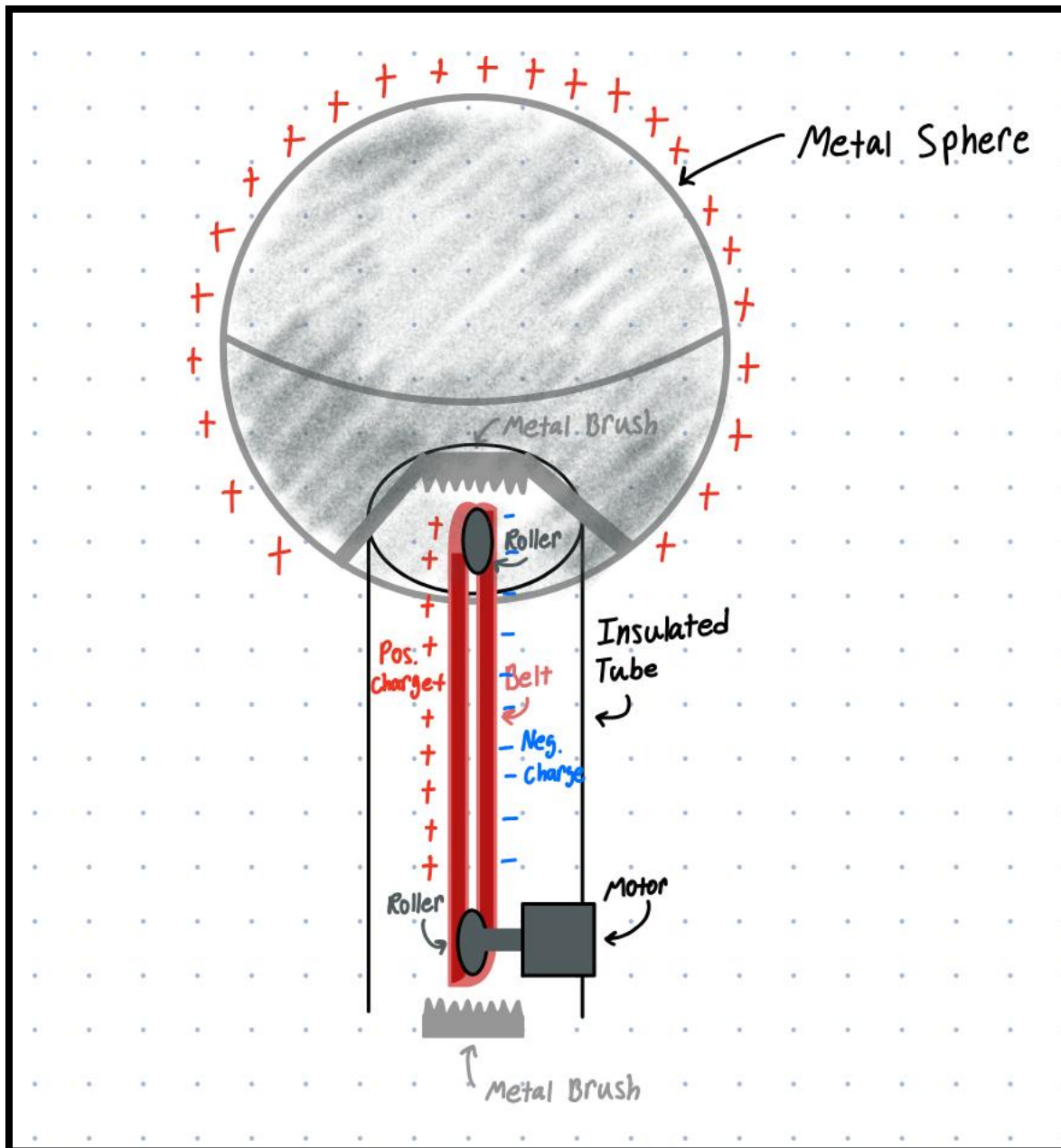


Figure 1: General Van de Graaff Generator Diagram

Although modern technology has advanced, the Van de Graaff generator still has its place in a few applications. One of which is its original purpose, nuclear physics. It is used to accelerate particles and ions for lower level of experiment investigating atomic properties. Modern day nuclear research requires far greater power than a Van de Graaf can provide though. The most significant use today is in semiconductor manufacturing. They are used to inject ions into

silicon, creating almost every piece of technology known today. Van de Graaff generators are also used in the medical field to observe how organic compounds react to particle radiation. They are also seen, of course, in many education settings such as a physics demonstration. This is because they are easy to build and effective at explaining multiple electrostatic properties with the various experiments that can be done with them. Each application of these generators is similar in how it is used to accelerate a particle or ion and send it into an object to utilize the reaction.

Although a Van de Graaff generator can produce millions of volts, usually a discharge is not lethal. It may have a high voltage, but the current is extremely small in comparison. The most that is likely to happen if a slight reflex of whatever created the spark. But even still, that would require a large Van de Graaff anyway. It may not necessarily be the most dangerous machine, but it is still important to be safe. Especially when the generators get to be on a much larger scale. If the generator is that large, it can produce enough current to be harmful. Environmental conditions have a large effect on the function of these machines. Humidity destroys the charge and makes it far less effective. That also means that the dryer the air is, the more charge can build, and a shock is more likely to happen. There are quite a few safety measures usually implemented. Grounding and nonflammable equipment is used to mitigate risk. PPE, like insulated gloves, suits, and mats, are typically used when operating these machines in a large-scale lab. In the event a shock was to occur, these measures should help prevent any actual harm.

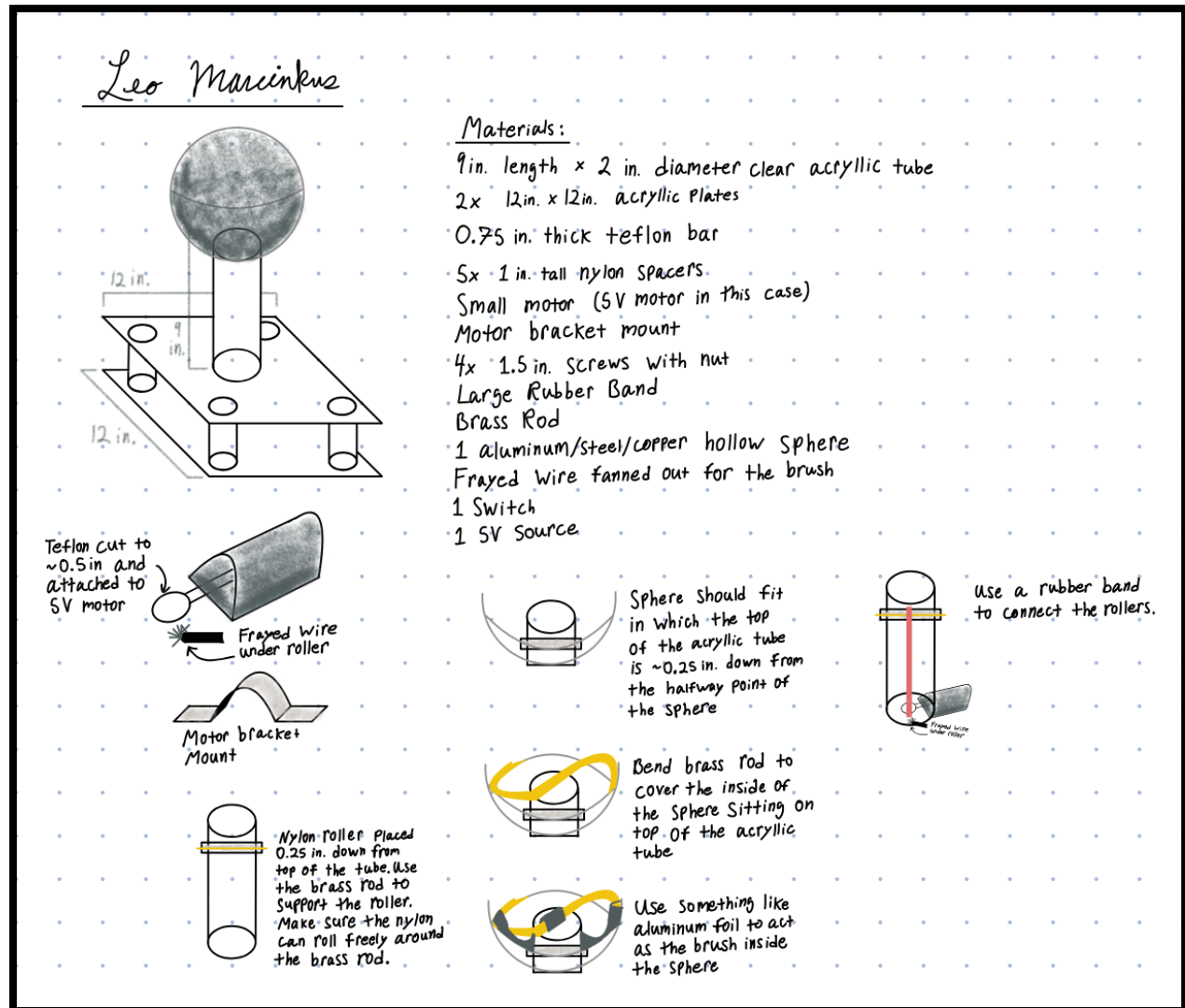
Design

Figure 2: Rough sketch of the design

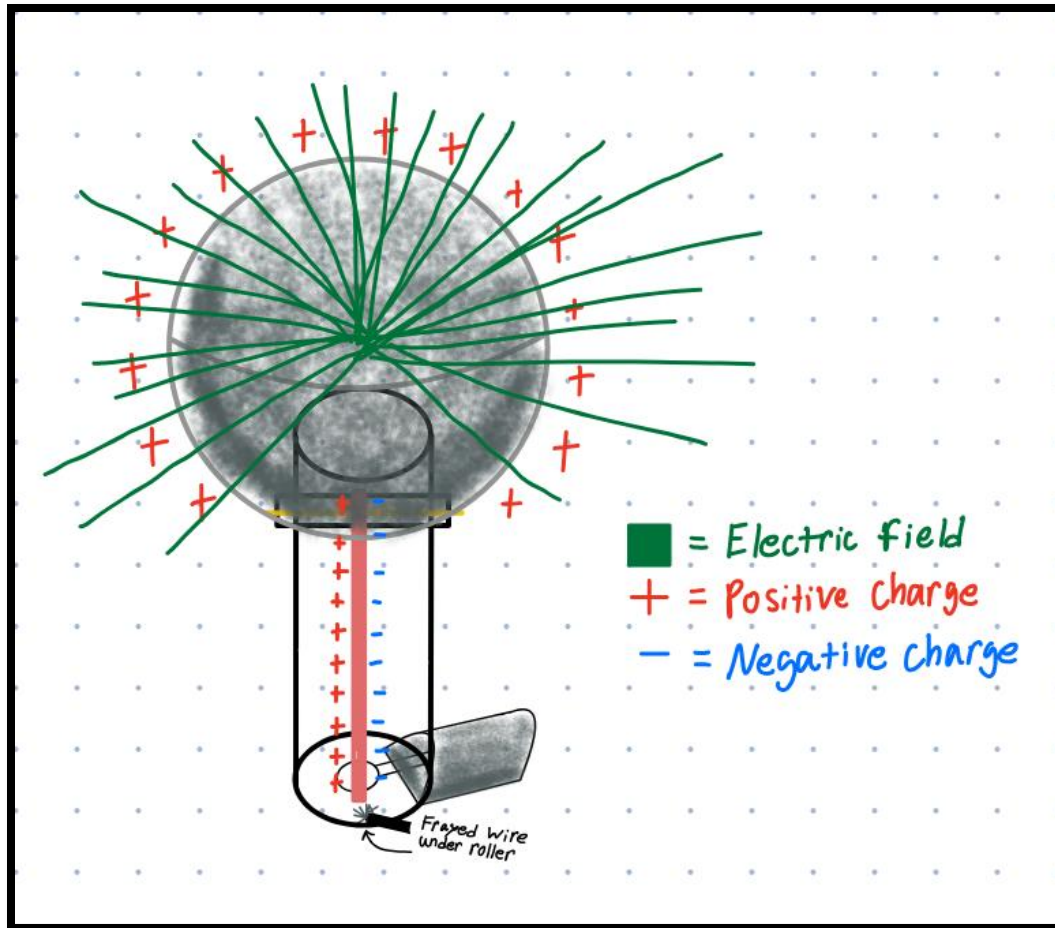


Figure 3: Electric field and charge of the Van de Graaff generator

The generator itself is not very complex. It distributes charge to a metal sphere using a rubber band moved by a motor. The first step to assembling the generator is to start with the two acrylic plates. Cut evenly spaced holes in the corners and use four of the nylon spacers to keep the plates separated. Use the 1.5" screws through the spacers to secure the boards together. A nut can be placed at the bottom of each screw once it is through the spacer. This will keep it in place. Wrap the bottom of the lower plate with aluminum tape or foil. Next, cut a hole in the middle of the top plate, about 0.75" to 1" will work. This is where the acrylic tube will be mounted allowing the rubber band to travel through. Mount the motor to the bottom plate using the motor mount. Ensure the motor's rod is centered with the drilled hole. Cut the Teflon rod so it is about

0.75" long and drill a hole in the center so it can be connected to the motor's rod. An adhesive might be needed to ensure the Teflon does not fly off the motor when it starts spinning. Now take the frayed wire and place it under the Teflon attached to the motor, but ensure they are not touching. It should be taped, screwed, or glued down so it does not move. This wire will function as the brush for the bottom of the rubber band.

Next, take the acrylic tube and 0.25" down from the top edge cut two holes. Take 1.5" of length of the brass rod and slide the last nylon roller over it. It should be able to spin freely. Place this in the holes that were just cut. Adhesive can be used for the edges to ensure the brass rod does not slide out and keeps the spacer/roller in place. Now take the acrylic tube and mount it above the hole that was drilled in the center of the top plate. Make sure the nylon roller is at the top of the tube, opposite the plate. Epoxy or glue can be used to ensure it bonds with the plate.

Take the bottom half of the metal sphere and cut a hole matching the acrylic tube. Slide the bottom half of the sphere down so the top of the acrylic tube is about 0.25" down from that seam of the sphere. A rubber ring or glue can be used to keep the bottom half of the sphere positioned whenever it is on the tube. Now take the brass rod and bend it in an "S" shape. Position the bent rod flat above the acrylic tube. It should contact the edges of the sphere. Now take aluminum foil/tape and wrap it around the brass rod just above the acrylic tube and to the sides of the sphere. Leave extra foil dangling down in the middle so it reaches just above the nylon roller and rubber band but ensure they do not come in contact.

With the bottom of the sphere assembled, it can be removed to attach the rubber band. Place the band around the top or bottom roller first and use a tool to reach into the tube and stretch it around the opposite roller. A bent metal coat hanger is useful to do this. Ensure the rubber band is taut and the rollers can still spin. Attach the top half of the sphere to the bottom

and place the sphere back on the acrylic tube. The last few steps involve the motor's power and switch. Any 5V power source can be used, but using batteries can make this small setup more portable. Mount the switch near an edge of the top plate, and the power source on the bottom plate near the motor. Solder the switch between the motor and its power source to ensure the generator can be turned off. Refer to the design images for better visualization for some parts of the assembly process.

The Van de Graaff generator is now built, but what can it output?

Mathematical Estimation of Maximum Charge and Potential

Electrostatic potential for a charged sphere:

$$V = \frac{kQ}{R} \rightarrow V = \frac{Q}{4\pi\epsilon_0 R}$$

Electric field for the surface of a charged sphere:

$$E = \frac{kQ}{R^2} \rightarrow E = \frac{Q}{4\pi\epsilon_0 R^2}$$

Assumption $E_{Breakdown}$ will be used:

$$E_{Breakdown} = 3 \times 10^6 \text{ V/m}$$

Assuming $E_{Surface} = E_{Breakdown}$:

$$\text{Maximum Charge } Q: E_{Breakdown} = \frac{Q}{4\pi\epsilon_0 R^2} \rightarrow Q = 4\pi\epsilon_0 R^2 E_{Breakdown}$$

$$\text{Maximum Potential in V: } V = RE_{Breakdown}$$

Theoretical maximum charge and potential:

$$Q = 4\pi(8.8541878128 \times 10^{-12} F/m)(0.05m)^2(3 \times 10^6 V/m) \approx 8.345 \times 10^{-7} C$$

$$V = (0.05m)(3 \times 10^6 V/m) \approx 150,000 V$$

Being more realistic would bring these values down significantly and improve my assumptions about the behavior of this design. The sphere used is not going to be perfectly smooth, as there will be a seam in the middle. The brushes used are not 100% efficient either. And the small 5V motor is only capable of producing a few μA , meaning it is not an especially powerful motor. A more realistic value can be calculated using β , a field enhancement factor, as follows:

$$E_{Local} = \beta E$$

$$V_{Realistic} = \frac{RE_{Breakdown}}{\beta} = \frac{(0.05m)(3 \times 10^6 V/m)}{3} = 50,000 V$$

The more imperfect the conditions for the Van de Graaff are the larger the value β would be. The most impactful limiting factors of the generator are the speed of the belt, air humidity, and any material buildup on the components affecting charge.

Demonstration

Demonstrating the capabilities of the Van de Graaff generator is a great way to understand how it works. A good demonstration for visualizing charge and potential can be taken from a common experiment with a separate smaller metal ball. PHYS 1120 here at CU even runs through this demonstration. While the generator is charging up, the instructor should attach a separate metal ball to a thread. Bring it close to the generator while holding the thread and letting the ball hang. The metal ball should first be attracted to the generator, once it makes contact, it should then be repelled. An effective way to illustrate the electric field breakdown in air is to use a wand. After letting the generator charge back up again, a metal wand can be brought out. It needs to be grounded for the demonstration to work. Once sufficiently charged, if moved close to the generator, a quick and loud spark should occur. It should even be visible if there is enough charge to be transferred. Now that the generator is discharged, aluminum pie plates can be placed on top. These illustrate the electric energy and conducting properties well. Allowing the generator to charge will eventually lead to each plate individually flying off the top of the generator.

With the metal ball, students should observe the initial attraction and then repulsion after contact. The main takeaway is that a charged conductor will induce the opposite charge to a nearby conductor. After contact, the ball receives the same potential as the generator's sphere and therefore gets repelled. This is great for providing an explanation of electric potential. When the wand is brought out, students should observe the spark that appears between the wand and the generator. The electric field near the generator's sphere increases until it overcomes the air's breakdown. This is known as dielectric breakdown. This demonstrates the electric field breakdown in air and is similar to a capacitor releasing its stored charge. For the pie plates,

students should see each pie plate in the stack jump off one after another. They conduct the charge from the surface of the generator and, because like charges repel each other, the pie plates fly off. The stored electric energy from the generator converts to lift the pie plates. This demonstration serves to understand electric energy and how conductors interact with each other.

Practice Problem

A small Van de Graaff generator exists with radius 5 cm. Assume the air around the generator has a breakdown field of $E_{\text{Break}} = 3.0 \times 10^6 \text{ V/m}$.

a). A grounded wand is brought close to the charged Van de Graaff generator. What is the surface charge density on the sphere? Find the theoretical maximum charge, Q_{Max} , the sphere can hold.

b). Using the theoretical maximum potential $V = RE_{\text{Break}}$ and thinking of it **like a parallel-plate capacitor** from part a), find the minimum gap between the generator's spherical surface and the grounded wand in which a spark will occur.

Solution

a).

$$\sigma_{Max} = \varepsilon_0 E_{Break} = (8.8541878128 \times 10^{-12})(3 \times 10^6 V/m) \approx 2.656 \times 10^{-5} C/m^2$$

$$Q_{Max} = \sigma_{Max} \times 4\pi \times R^2 = (2.656 \times 10^{-5} C/m^2)(4\pi)(0.05m)^2 \approx 8.345 \times 10^{-7}$$

b).

$$V = RE_{Breakdown} = (0.05m)(3 \times 10^6 V/m) = 150,000 V$$

$$Parrallel Plate Capactor Field \rightarrow E = \frac{V}{d}$$

$$E_{Break} = \frac{V}{Gap_{min}} \rightarrow Gap_{min} = \frac{V}{E_{Break}} = \frac{150,000 V}{3 \times 10^6 V/m} = 0.05m$$

Citations:

Khalid, R., S. Shaheen, Aqeel-Ur-Rehman, and A. Javed. “Van de Graaff Generator – A Cost Effective Solution.” *2018 International Conference on Power Generation Systems and Renewable Energy Technologies (PGSRET)*, 2018, pp. 1–6. IEEE, <https://doi.org/10.1109/PGSRET.2018.8685983>.

Plasma Channel. “*DIY 40000 Volt Van Der Graaff (Battery Powered)*.” YouTube, 7 Dec. 2018, www.youtube.com/watch?v=b-Gs81_-_aU